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Critical review on water quality analysis using IoT and machine learning models

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ABSTRACT

Water quality and its management are the most precise concerns confronting humanity globally. This article evaluates the various sensors used for water quality monitoring and focuses on the water quality index considering the multiple physical, chemical, and biological parameters. A Review of Internet of Things (IoT) research for water quality monitoring and analysis, sensors used for water quality can help remote monitoring of the water quality parameters using various IoT-based sensors that convey the assembled estimations utilizing Low-Power Wide Area Network innovations. Overall, the IoT system was 95 % accurate in measuring pH, Turbidity, TDS, and Temperature, while the traditional method was only 85 % accurate. Also, this study reviewed the different A.I. techniques used to assess water quality, including conventional machine learning techniques, Support Vector Machines, Deep Neural Networks, and K-nearest neighbors. Compared to traditional methods, machine learning and deep learning can significantly increase the accuracy of measurements of groundwater quality. However, various variables, such as the caliber of the training data, the water quality metrics' complexity, and the monitoring frequency, will affect the accuracy. The geographical information system (GIS) is used for spatial data analysis and managing water resources. The quality of its data is also reviewed in the paper. Based on these analyses, the study has forecasted the future sensors, Geospatial Technology, and machine learning techniques for water quality monitoring and analysis.

Introduction

Water is a highly significant need of every single living being. With an increase in the human population, the requirement for preserving water assets is acquiring greater significance. Several parameters are used for identifying water quality based on the samples' water level, pH level, turbidity, and salt content. The machine learning techniques are combined with a GIS based on data availability and quality. Machine learning algorithms must account for spatial autocorrelation and temporal dynamics to predict water quality. Incorporating spatial analysis techniques within GIS, such as geostatistics, can help capture these variations and improve the accuracy of predictions. By addressing these problems, robust machine learning and GIS-based frameworks can be determined for water quality index calculation, aiding in efficient water management and decision-making processes. Water quality analysis plays a vital role in assessing and monitoring the health and safety of water sources. It involves evaluating various physical, chemical, and biological parameters to determine whether the water quality meets the required standards for its intended use, such as drinking purposes, recreational activities, and ecosystem preservation. Traditional water quality monitoring systems have relied on conventional laboratory-based methods in which water samples are collected from various resources and transported to the laboratory for analysis. Water samples are collected manually from different points of interest, such as rivers, lakes, and wells, and detailed studies are performed using specialized equipment and techniques in laboratory conditions. The obtained data is analyzed and interpreted to assess the water quality. It is compared against the established standards or guidelines to meet the criteria for its designated purpose. The limitations of traditional laboratory-based water quality monitoring systems are that they are time-consuming

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and costly, lack real-time monitoring, and are inaccessible to remote locations.

Data-driven water quality assessment

Water is a fundamental resource that has always supported different human activities, including farming, industry, and homegrown use. However, the nature of water assets is progressively compromised by anthropogenic activities, such as contamination from modern releases, farming overflow, and urbanization. Checking and evaluating water quality is essential for ensuring the supportability of water assets and safeguarding human well-being. Traditional methods of water quality assessment deal with manual sampling and laboratory analysis. It is a time-consuming process.

The advent of data-driven technologies, including sensors, the Internet of Things (IoT), A.I., and Geographic information systems (GIS), has changed the field of water quality evaluation. Information-driven approaches offer a few benefits over conventional techniques, including Real-time monitoring, High-resolution data, spatial and temporal analysis of water quality, and predictive modeling. The data-driven water quality assessment is essential for addressing the following challenges: Increasing water scarcity, pollution from nonpoint sources, and the impact of climate change. Data-driven water quality assessments to ensure the availability of clean and safe water for future generations. The framework of water quality data analysis proposed by WHO and BIS.

This literature review explores the potential of data-driven approaches in accurately predicting water quality index (WQI) using machine learning and GIS techniques. The study will focus on the following key aspects:

1. Machine learning algorithms: Identify and evaluate the most effective machine learning algorithms for WQI prediction.
2. GIS integration: Explore the role of GIS in spatial analysis and visualization of WQI data.
3. Data considerations: Address the challenges and opportunities associated with data collection, preprocessing, and management for data-driven WQI prediction.
4. Performance evaluation: Discuss the metrics and methodologies for evaluating the performance of data-driven WQI prediction models.
5. Case studies: Analyze real-world case studies to demonstrate the practical application of data-driven WQI prediction in various water resource management scenarios.

Review of analysis of water quality parameters and indexes

Various physio-chemical and biological parameters for measuring and monitoring the Water quality index (WQI) exist worldwide. The assessment procedure of WQI is analyzed and summarized in this section. Analyses of various water quality measurement parameters have been made using traditional methods that help to calibrate the physical changes. (Wu et al., 2021) Carried out experimental work on the critical parameters that affect water quality. Fifteen physio-chemical parameters were evaluated to assess the WQI and categorized into three groups. These helped establish the results with minimum error and good accuracy. This detection benefits low-cost and fast-track water quality evaluation in adjacent basins. Based on the statistical values, the spatially evaluated distribution pattern and different models of WQI are classified. Given the expansive implications of water quality prediction utilizing a complex methodology, our review aims to address fundamental research questions, making significant contributions to the field. The recognized research questions directing our examination are as per the following (Kar & Varsha, 2023):

R1: How do integrating Internet of Things (IoT), AI, and Geographic Data system (GIS) functionalities add to the headway of water quality expectations?

R2: What strategies can be figured out by associations to develop robust water quality forecasts using machine learning models that align with sustainability objectives?

The answer to the above questions is that the present study Focuses on the intersection of IoT, A.I., and GIS in water quality expectations. This review surveyed 67 articles from driving diaries in the Scopus data set (Sharma et al., 2021). This strategic methodology permits us to introduce thorough bits of knowledge and exploratory discoveries. Sections 2 and 3 focus on literature reviews and comparison of various Review methodologies. Section 4 illustrates the findings of the systematic literature review on GIS, and Section 5 discusses the machine learning algorithms and multiple techniques implemented in various applications in the research area. Section 6 is the conclusion.

(Uddin et al., 2021) have developed the traditional methods of WQI models. Due to its generalized structure, the evaluation cites the conversion of substantial data on water quality into a single value or index as one of the valuable techniques. They divided WQI models into four consecutive steps: selection of parameters of water sub-indices generation, parameters weighting values, and overall WQI calculation. Therefore, the large volume of water quality data is compressed into a single index with a comparison between various traditional WQI Index related to selecting parameters, sub-index procedure, weighting, aggregation techniques, and rating scale. (Nayak et al., 2020) analyzed the water quality status of Indian rivers.

(Bui et al., 2020) Assessed WQI and concluded that traditional WQI calculations consumed more time and identified few errors by calculating sub-indices. Numerous statistical and visual evaluation indicators were used to evaluate the models. The machine learning algorithm divided the data into training and testing sets, used hybrid algorithms, and estimated WQI Values. (Wu et al., 2017) to obtain WQI analyses, the research work was carried out in Lake Poyang, China, where 24 samples of water quality values were classified into three groups. This work considered 20 different water quality parameters for analyzing Total Nitrogen (TN) and Total Phosphorus (TP), whereas hazardous metals and other criteria were rated low in the WQI Analysis. (Wang et al., 2017) Worked with a broad classification of different fractional deviation methods referred to difference, ratio, and normalized difference index. The obtained WQI values vary from 56.61 to 2886.51. The above index was estimated using the slope of the curve and the root means square error values.

(Lamare & Singh, 2016) Did the work on Numerus's pollution index (NPI), the simplest WQI evaluation method known among all the WQI methods. Their study evaluated the specific parameters for their potential contribution to water bodies. (Dohare et al., 2014) Examined water quality for Indore City, and they calculated WQI using around 27 parameters: pH, color, Total Dissolved Solids (TDS), calcium, chromium, electrical conductivity (E.C.), total alkalinity, hardness, zinc, manganese, and nickel. The evaluated results are as per IS 10500–2012 specifications. They used statistical methods to analyze the physio-chemical parameters and water pollutants characterization.

(GG.S, 2013) A standard formula for WQI was used by fixing the numeric Q-value and weightage factor. Despite fixing the weight factors for several parameters, they adopted the dissolved oxygen level, fecal coliform, B.O.D., pH, nitrate, Phosphorus, temperature, turbidity, and TDS Using these conditions, they arrived at an equation to calculate WQI with missing parameters. (Nayar, 2020) Analyzed Physio-Chemical parameters: temperature, Transparency, hardness, pH, sulfate, chloride, D.O., B.O.D., C.O.D., alkalinity nitrates phosphates, and water quality. They classified parameters as (0–25) - Excellent quality, (26–50) - Good quality, (51–75) - poor quality, (76–100) Inferior quality, >100 unsuitable for drinking water.

Finally, (Uddin et al., 2021) employed the Analytic Hierarchy Process (A.H.P.) to categorize the World Quality Index as exceptional (90–100), good (90–75), fair (75–50), marginal (50–25), and low (25–5). This proves to be a very efficient method for minimizing uncertainty, weighting procedures, and improving accuracy. Table 1

Table 1
Comparison of WQI Models weighted parameters used by various researchers.

Parameters	Horton (Jha et al., 2015)	National Sanitation Foundation (Dawood et al., 2022)	W.J.WQI (Uddin et al., 2021; Liu et al., 2019)	Environmental quality index (Uddin et al., 2021)	Scottish Research Development Department Index (Sutadian et al., 2016)	Bascarron Index (BWQI) (Sun et al., 2016)	Said (Uddin et al., 2021)	Oregon (Swamee & Tyagi, 2000)
pH	4	0.11	–	0.1	0.09	1	–	0.16
Turbidity		0.08	–	0.1	–	4	0.1	–
D.O.	4	0.17	0.10	0.1	0.18	4	1.5	0.16
NO_3		0.10	0.06	0.15	–	2	–	0.16
Cl	1	–	0.07	0.1	–	1	–	–
Temp	–	0.10	0.03	0.1	0.05	1		–
T.S.	–	0.07	–	0.1	–	–	–	–
B.O.D.	–	0.11	–	0.1	0.15	3	–	0.16
Colour	–		–	0.1	–	2	–	
F.C.	1	0.16	0.17	0.1	0.12	–	15	0.16

presents details of the different kinds of WQI models and weightage used by various researchers. Several authors have analyzed some parameters. A comparative analysis by different authors shows that pH analysis of the Horton index shows better results than any other index method. The Bascarron index shows better results for calculating turbidity, and other remaining parameters showed better outcomes for Bascarron index-based weightage results. A cluster analysis of water quality spatial distribution given by various research studies is presented in Table 2.

Review methodology

There are several proposals for a writing survey. It involves a four-step approach for doing the writing survey, as depicted in the ensuing parts. This technique helps us recognize and concentrate on topics and future exploration headings. Fig. 1 shows the flowchart of the review methodology adopted for Water quality prediction using Sensors, IoT, A. I., and GIS technologies.

Review of water quality analysis using GIS

The WQI can be predicted by plotting the spatial characteristics of water in lakes (Wang et al., 2017). Some of the parameters were estimated using remote sensing monitoring Technology. Depending on the problem domain, the boxplot analysis provides an accurate visualization prediction and finalizes the outlier detection of threshold values. Machine learning techniques were applied based on the captured image data sets from the Gravity Recovery and Climate Experiments (GRACE). These helped with accurate water quality predictions and images of groundwater distribution.

(Nampak et al., 2014) A GIS approach was utilized to incorporate five contributing variables: lithology, drainage, land cover/land use, lineaments, seepage, and slope. The outcomes showed that around 1.2% of the review region was assigned as having great potential for groundwater to re-energize (Yeh et al., 2016). The curve between the wavelength and reflectance of the WQI value for different sample sites is discussed. Analytical hierarchy-based images are used to identify suitable solid waste dumps based on the assigned weights of the various parameters. The map was prepared to classify the trustworthy dump site as highly qualified, less fortunate, and unsuitable for the site selection. So, the multi-criteria decision-making analysis was done to identify the location, presented in Fig. 2.

Researchers and water resource professionals effectively integrate an GIS and visualize the various groundwater data layers by employing a methodological approach for groundwater analysis using GIS This approach enables a better understanding of the spatial patterns, trends, and relationships between groundwater parameters and environmental factors. The integration of GIS with hydrological modeling and decision support systems facilitates informed decision-making, sustainable groundwater management, and protection of valuable resources. The methodology describes the data acquisition system using wired and wireless sensors. Hydrological modeling tools simulate the groundwater flow and evaluate the water availability. Various data sets are pre-processed, and model selection, training, and evaluation of multiple metrics are predicted. Some examples where the contexts based on GIS are discussed for providing ideas about spatial and temporal mapping along with the implementation of A.I. models are listed in Table 3.

Review of sensors-based water quality monitoring systems

Continuous monitoring of water quality parameters can be measured

Table 2
Water Quality Parameters Utilized by Different Researchers.

Author Reference							
Water parameters	(El Bilali & Taleb, 2020)	(Jha et al., 2015)	(Dohare et al., 2014)	(Ewaid et al., 2019)	(Ewaid et al., 2018)	(Chen et al., 2020)	(Shakhari & Banerjee, 2019)
pH	7.90	7.567	7.70	7.909	7.43	–	7.979
Conductivity (μS)	–	585.3	–	–	–	–	–
Water temperature	20.1	16.13	–	33.00	–	–	–
Turbidity	–	38.35	40.36	63.87	–	–	–
B.O.D. (mg L^{-1})	–	–	9.381	2.98	2.23	–	7.80
Dissolved oxygen (mg L^{-1})	8.69	–	5.67	–	–	–	10.4
Total Nitrogen (mg L^{-1})	3.95	–	6.50	–	3.47	–	7.06
Ammonia (mg L^{-1})	0.80	–	–	–	–	–	
Nitrate (mg L^{-1})	1.13	–	–	–	–	–	–
Total phosphorous (mg L^{-1})	0.27	–	0.058	–	–	–	1.7
Calcium (mg L^{-1})	23.8	–	–	224	92.0	–	–
Chloride (mg L^{-1})	23.7	–	–	–	–	–	883.8
Sulphate (mg L^{-1})	22.8	–	91.36	–	358.15	–	–

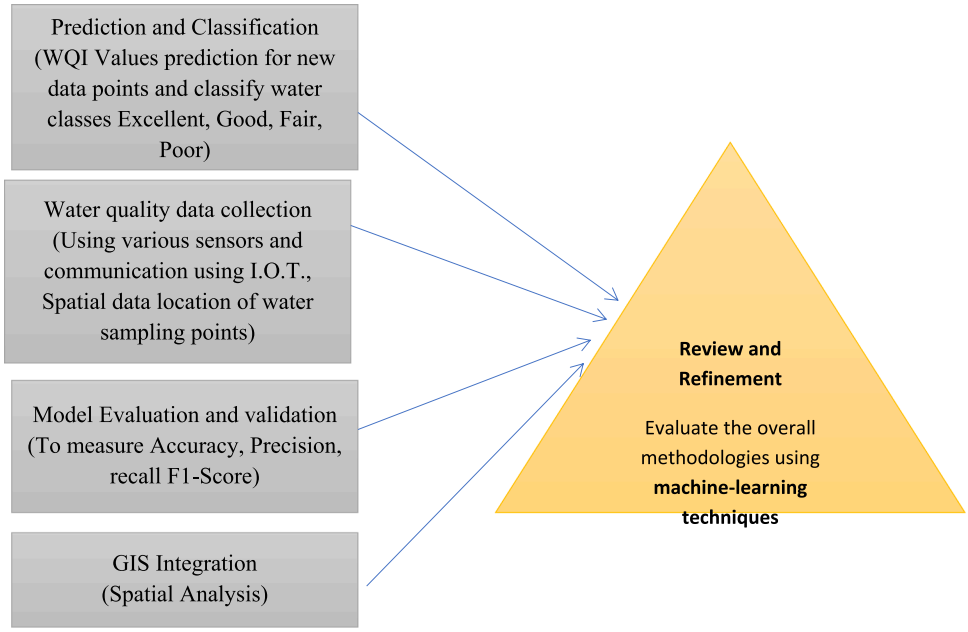

Fig. 1. Flowchart on review methodology using various approaches for water quality prediction (Kar & Varsha, 2023).

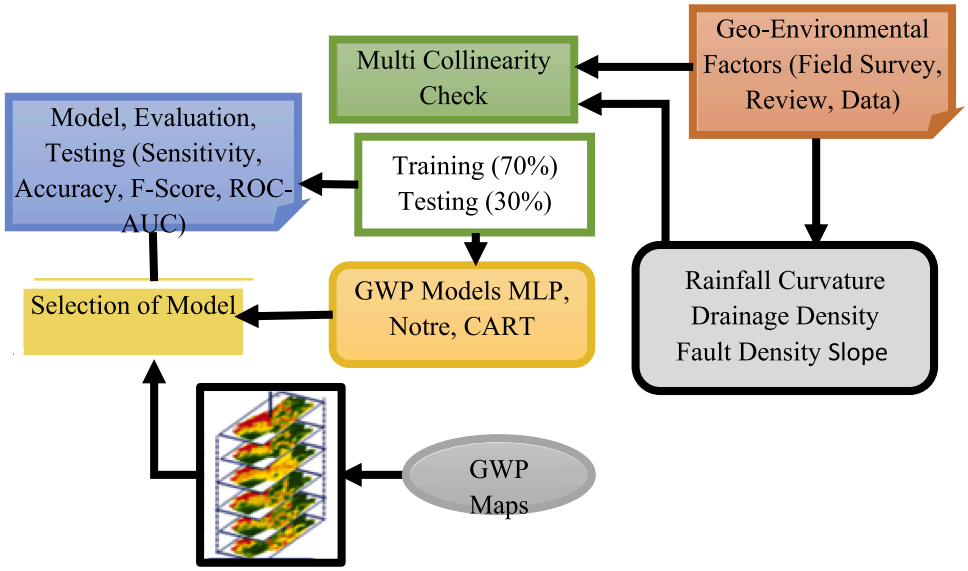

Fig. 2. Methodological approach for groundwater analysis using GIS (Arabameri et al., 2021).

using different portable and low-cost approaches. (Tuna et al., 2013) Focused their work on the practical usage of mini boat probes loaded along with the wireless sensor network monitoring system (WW.S.N.. IEEE 802.15.4 is a wireless interface system meant for continuous water quality monitoring, and a few parameters, including DD.O. pH, EE.C. temperature, turbidity, and nitrate, to determine water quality, and simulation studies were focused on monitoring the water quality of drinking water reservoirs. (Wu et al., 2021) analyzed the maximum water quality parameters meant for better accuracy. (EE.L.Watershed et al., 2017) Analyzed a few parameters to get the nearest accurate values.

The system integrated with GG.P.S. provides an online data acquisition system with two long-range radio modems for simultaneous data transmission. The MATLAB-based simulation environment was also used for analyzing the water quality parameters with the source codes of MM. A.T.S.N.L.(MM.A.T.L.A.B. Wireless Sensor Node Platform Lifetime Prediction & Simulation Package). The lifetime of node and transmission power was analyzed, along with variations in the transmission power of the nodes between 0 dB and −25 dB. Fig. 3 shows the wireless sensor network-based monitoring system. A field test in the water reservoir was done in Kirklareli Dam. A web-based online interface with remote monitoring IoT devices controlled the system. Researchers were focused on the parameters such as D.O. pH, EE.C. temperature, turbidity, and nitrate. They concluded the simulation results using a web-based interface and the system's dependence on parameters such as transmission frequency, power, size of the packet node parameters, etc.

It is important to note that the sensors monitoring the water quality should be cost-effective and continuous. The micro sensors fabricated along with the channel interface should measure water quality parameters such as pH and conductivity. The final results were compared with the commercial sensors in static and dynamic conditions. Fig. 3 shows the resistance of the pH sensor measured in static conditions. The same

Table 3
survey study on GIS and fitness in water quality assessments (Venkatachalam & Ray, 2022).

Year	Authors	Review	Context discussed
2022	(Dawood et al., 2022)	Application of Water Quality Index and Multivariate Statistical Techniques to Assess and Predict Groundwater Quality with the Aid of Geographic Information System	This study assessed and mapped groundwater quality and spatial distribution. Multivariate statistical techniques were used to identify sources of pollution and develop a model to predict water quality index.
2019	(Thinsungnoen, 2019)	An Integrated System of Geographic Information and Water Quality: Lamtakong River	This study developed a decision tree model to predict water quality based on land-use factors and created a GIS tool for retrieving and displaying water quality data
2018	(Tiwari et al., 2018)	Understanding the spatial and temporal dependence of the migration of conservative contaminant plumes in the urban groundwater environment in the Panchkula region, Haryana, India	Analyzing the plume's movement serves as a proactive step, laying the groundwork to effectively mitigate potential contamination in the areas surrounding the disposal site.
2006	(Ghose et al., 2006)	GGISbased transportation model for solid waste disposal – A case study on Asansol municipality	A proposed GIS optimal routing model aims to identify the most cost-effective and efficient collection routes for transporting solid wastes to landfills.

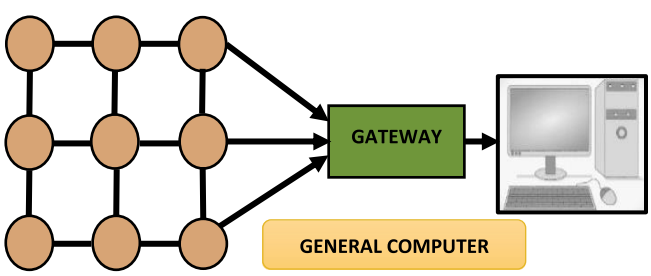

Fig. 3. W.S.N. based on water quality monitoring sensors (Bell, 2013).

sensor is mounted with different pH liquids can find a change in the pH value regardless of the exposure time. IoT-based water quality measuring system using sensor-based circuitry powered by solar renewable energy analyzed pH level, turbidity, salinity, arsenic level, etc. The information collected from the sensors was collected in cloud computing using a Wi-Fi module, and the graphical comprehension demonstrated data via Think Speak.

(Azpilicueta et al., 2022) Applications for monitoring subsurface water quality have grown in popularity in recent years in low-power wide area networks (L.P.W.A.N.s). For this application, L.P.W.A.N. provides several benefits, including Long-distance communication for monitoring groundwater quality. L.P.W.A.N. devices offer low power consumption and help monitor remote applications where replacing the batteries regularly and running for extended periods is impractical. L.P.W.A.N.s is cost-effective for monitoring the quality of subsurface water, and several research works have investigated the use of L.P.W.A.N.s for monitoring the quality of underground water. A variety of water resources can be monitored using L.P.W.A.N.s and Numerous aspects of water quality, such as pH, temperature, turbidity, and conductivity, can be tracked using L.P.W.A.N.s Real-time water quality monitoring through the deployment of L.P.W.A.N. scan assist in identifying and responding in polluted situations. Overall, the research on LL.P.W.A.N.s for monitoring the quality of subsurface water still encourages the application of LL.P.W.A.N.s has a variety of benefits, and they are being applied more frequently.

(Nayar, 2020) Used wireless sensor network technology with the GSM Module and ZigBee protocol to create a low-cost water quality monitoring system. The microcontroller of the electrochemical sensor was used to create a hardware system with wireless communication networks for monitoring water temperature, dissolved oxygen, and pH values in a pre-programmed period. (Bell, 2013) Authors have employed a wide range of sensors to measure pH, turbidity, water level, temperature, and humidity of water. They used a pH sensor, Turbidity sensor, Ultrasonic sensor, DHT-11 sensor, and microcontroller Node MCU with E.S.P.Wi-Fi module, think to speak IoT data collection application. (Sanya et al., 2022; Bates et al., 2021). This study uses LoRaWAN to monitor the water quality in rivers and lakes. The authors emphasize LoRaWAN's benefits in terms of long-distance connectivity, low power consumption, and affordability. They discuss the installation of LoRaWAN sensors to gauge variables in water bodies, including pH, temperature, and dissolved oxygen. The study proves the viability and dependability of LoRaWAN for monitoring water quality in real-time applications, and the authors discussed the choices for data transmission, network architecture, and sensor selection. In terms of data accuracy, transmission range, and power consumption, they assess the system's performance, and the outcomes show how the LoRaWAN-based system is accurate and successful in monitoring the water quality metrics. Fig. 4 Shows the graph for the sensor ranges adopted by the different researchers to monitor pH values.

(Almetwally et al., 2020) designed and developed a system for automatic decision-making to control water quality parameters. The different water quality sensors resulted in acceptable limits and recorded historical readings in a cloud platform. The system is related to public health and eliminates the inconvenience faced during offline lab analysis of collected samples. (Kothari et al., 2021) Authors have used a variety of sensors, including temperature, TDS pH, and DD.O. sensors to measure the WQI parameters. The information gathered from the GSM SIM 900 module and Arduino mega2560 microcontroller communicated the information to the data set server at Things Peak. The information sent was seen utilizing an Android IoT application associated with the server and the. The information showed the effective continuous application of getting the clients ready to check the sensors on an Android versatile screen. (Roseela et al., 2020) Implemented submerged remote correspondence accomplished through IoT made a trend-setting innovation with the microcontroller and Wi-Fi. In this review, temperature, turbidity, and distance between the articles were estimated using respective sensors associated with the transmitter and recorded at the recipient side, utilizing IoT innovation for diverse time stretches (Akbarighatar et al., 2023), [57]. The proposed U.W.C. framework was seen as dependent on IoT and known for its minimal expense and efficiency, and it can be utilized for various applications. The sensors used

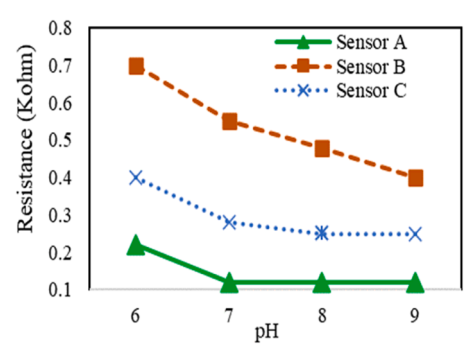

Fig. 4. pH Resistance under static conditions.

for various applications for measuring water quality play an essential role. Table 4 details various authors' sensors for measuring water quality and suitable hardware, software, and networking systems locations to ensure efficient and accurate output.

Water quality analysis using machine learning techniques

Employing IoT (Internet of Things) and machine learning for water quality monitoring offers advanced capabilities for collecting, analyzing, and interpreting data. (Wang et al., 2017) proposed a machine learning technique employing WQI and fractional derivative methods to get remote sensing spectral indices. A support vector Regression model was implemented to analyse the model's accuracy. (Liu et al., 2019) Work on smart water quality analysis and the prediction was done using Long short-term memory (LSTM) time series neural network IoT Environment. They used training and test monitoring data to analyse the water quality metrics to make predictions. The final results demonstrated a good indication of the model's anticipated and actual values using multidimensional input data sets and assessment for the future emerging water quality trend using LSTM deep neural networks to forecast drinking water quality.

(Shakhari & Banerjee, 2019) The authors used a multi-class classification system with continuous data sets to regulate water quality. They efficiently analyzed data using an algorithm known as the decision tree classification model to forecast the water quality. The research is employed in water quality monitoring equipment to classify the quality of bottled drinking water, pipeline drinking water, and pond water taken at center stage. They compared the quality of two methods of classification and regression. (Sharma et al., 2021) A supervised algorithm and machine learning approach accomplished the water quality prediction. This approach needed to be updated with drawbacks, such as lab time consumption, cost, and statistical analysis. Analysis of four different parameters, namely turbidity, temperature, pH, and TDS, shows that polynomial regression with a level of 2, angle assistance, and learning rate of 0.1 outperformed other relapse computations by its proficiency in most accurate WQI anticipation.

(Lamare & Singh, 2016) did the prediction of water quality for irrigation purposes using six models of machine learning techniques exhibited A.I. models as proficient methods for the precise view of the nature of water systems. Robotized sensor implementation and machine learning models help researchers regulate water quality and address irrigation water quality issues. An automated sensor technology coupled with K-NN and S.V.R. machine learning provides accurate results. (Ewaid et al., 2018) performed groundwater quality prediction using machine learning techniques, and the initial step involved in the application of regression approach to predict groundwater images and the process adopted are shown in Fig. 5. Gaussian Mixture Model was used to produce minor errors, and the machine learning models were applied in the GRACE Dataset to predict future datasets in the form of pixels.

Data mining algorithms with the combination of standalone and

Table 4
Review of Hardware, sensor, and I.O.T. techniques used by the researchers.

Sl. No	Author Reference	Sensors and Hardware tools used	Parameter measured	Accuracy level
1	(Kar & Varsha, 2023)	Smart water quality sensor	pH, D.O. conductivity, temperature, oxidation–reduction potential, turbidity, dissolved ions	The mean value of 69.01 Accuracy level:93.08 %
2	(Wang et al., 2017)	Optical sensors, Microelectronics Mechanical systems, and Bio-Sensors	pH, D.O. Turbidity, Escherichia Coli, and Residual Chlorine.	Accuracy level:87 %
3	(Azpilicueta et al., 2022)	Kapta 3000 AC4 sensor Spectro: laser	Chlorine, Temperature, pressure, conductivity, Turbidity, and color,	–
4	(Najah Ahmed et al., 2019)	The system combines computer image processing technology with biological sensors.	Metal toxic pollutants Hg2+, Cu2+ and Zn2+	Accuracy level:93.3 %
5	(Nayar, 2020)	DD.O.and pH sensors and zhorrible water checker, WW.S.N. GSM Technology	D.O. pH values	Accuracy level:96.3 %
6	(EE.L. Watershed et al., 2017)	SCADA, internet in Raspberry Pi	Collection, storage, analysis, and visualization	SCADA Accuracy-94 %
7	(Tuna et al., 2013)	Temperature sensor, pH, Turbidity, Ultrasonic sensor, Arduino Uno, water flow sensor, Filter unit	pH, turbidity, temperature	–
8	(Demetillo et al., 2019)	Data logging system with a 32-bit AA.R.M.core microcontroller board based on the Atmel SAM3 × 8E ARM Cortex-M3	pH, T	TDS Data logger Accuracy-97 %
9	(Liu et al., 2019)	Arduino and Node MCU, PP.H. Sensors, Turbidity sensor, Dallas Temperature sensor, DD.O. sensor, salinity sensor	pH, Turbidity, Temperature, Dissolved Oxygen, and E	E.C. Accuracy of sensors-91.2 %

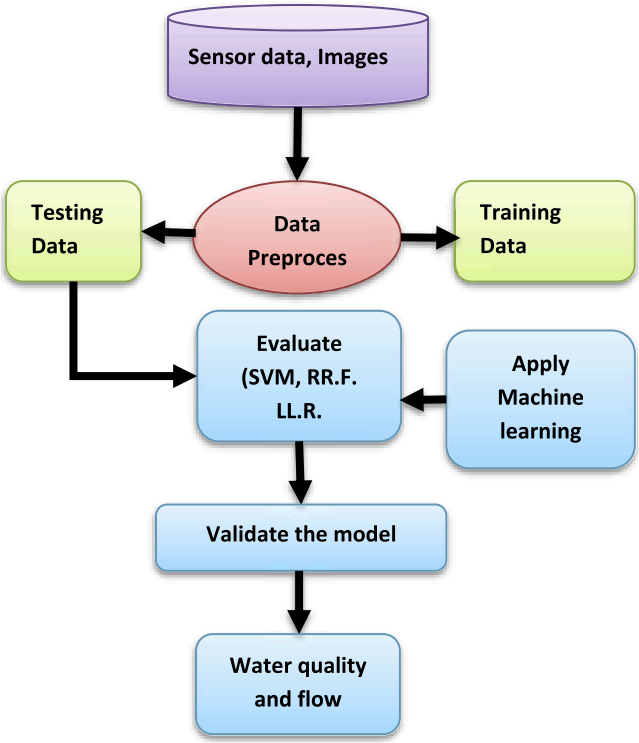

Fig. 5. Flow chart of an implementation of machine learning models (Hussein et al., 2020).

hybrid algorithms models were evaluated. Faecal coliform (FF.C. and total solids (TT.S. are the two parameters considered and revealed. (Mancuso et al., 2021) used post-processing techniques produced by time series as a forecasting method, and this time series with hierarchical extracts showed the end Neural Network with an accurate forecast. The Neural Network disaggregation model was created, tested, and trained by different data sets. (Dilmi & Ladjal, 2021) used the Long Short Term Memory deep learning approach to classify the water quality network of recurrent neurons and achieved an accuracy of 99.72% using models for time series prediction, with popularity in deep learning techniques. They used a multisensory monitoring system for the construction of the database. The performance assessment was done using AI techniques, LSTM and RNNs, and multi-class SVM. Different types of analysis were out for data reduction. The technology used principal component analysis (P.C.A., linear discriminant analysis (LDA), and independent component analysis (II.C.A..)

(Chen et al., 2020) Did a comparative analysis of water quality prediction using machine learning with extensive data analysis?. The comparison was made between 10 learning models for assessing the quality performance using big data. (Herath & Mittal, 2022) (Yang et al., 2019) The data sets collected are related to the parameters pH, D.O. C.O. D. Mn, and NH3 and validated by the model's Decision tree (D.T., Random Forest(RR.F. and Deep cascade Forest(D.C.F.. The best models were identified based on the increased efficiency in predicting surface water quality. (Aldhyani et al., 2020) Carried out work on water quality prediction using AI Algorithms that includes nonlinear autoregressive neural network (N.A.R.N.E.T. and long short-term memory (L.S.T.M. deep learning algorithm. Seven datasets using the L.S.T.M. algorithm are shown in Fig. 6. Table 5 analyses the various machine learning algorithms used in water quality analysis by multiple researchers.

Using an artificial neural network and regression plot, the comparison of SVM, K.N.N. Nave Bayes models were made, and the output from test data was found to improve the accuracy in the prediction of WW.Q.I (Uddin et al., 2021) The authors analyzed the WQI and WQC using machine learning techniques and proved that the GARNET result demonstration provides a better forecast than LSTM Following that (Dohare et al., 2014) is the prediction of WQI based on hybrid data mining algorithm. Based on the above techniques, their approach may consume time and lead to errors and deviations in sub-indices—Standalone (4 algorithms) and hybrid data mining (12 algorithms). Hybrid

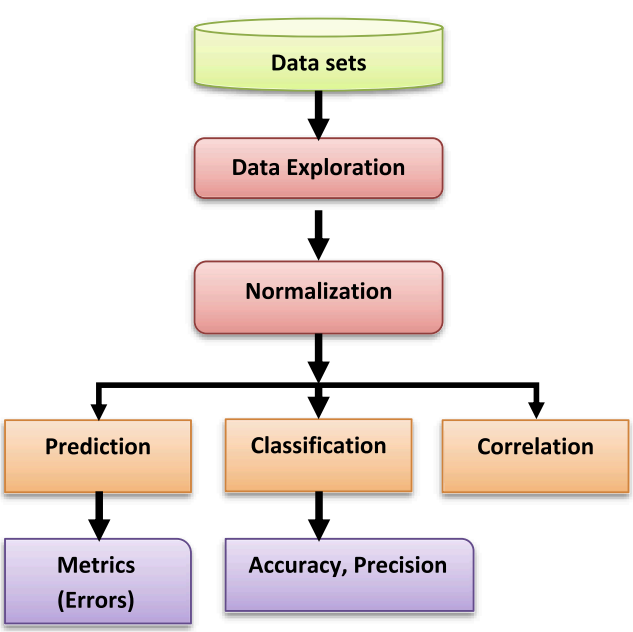

Fig. 6. Flow diagram for estimating efficient algorithm (Uddin et al., 2021).

bagging (B.A. and Random tree (RR.T. outperformed the other models, and they used a boxplot to predict the errors such as R^2, RMSE. MAE, and NSE. (GG.S, 2013) used an algorithm that works with big data analysis with a reduced cost to improve the prediction efficiency. They observed D.T. R.F. and D.C.F. give better predictions than other algorithms. (Venkatachalam & Ray, 2022) The trained requirement is necessary to train the algorithm to be more accurate. (Ahmed et al., 2019) They analyzed the data using a supervised algorithm and considered the error and deviations for predicting WQI and WQC He concluded the gradient boosting algorithm is more efficient than other regression algorithms in estimating MAE. MSE. RMSE and R^2. and the polynomial regression shows better results compared to others. In the present study, the authors mainly focus on the WQI and WQC. parameters to predict the future performance and efficiency of different algorithms. This research primarily focuses on comprehending the challenges related to the utilization and consequences of the emerging generation of AI-based systems in decision-making processes. This work's objective involved identifying the potential research opportunities specifically for Information Systems (II.S.. Water quality monitoring can be done in real-time using data analytics. This can aid in the early detection of potential issues so that they can be fixed before doing significant harm (Pasika & Gandla, 2020). For instance, data analytics can be used to monitor the concentrations of contaminants like microorganisms, heavy metals, and chemicals in water. These details can be used to pinpoint regions where there is a chance of poor water quality and to take preventative measures.

(Herath & Mittal, 2022) Analyzed high-impact impacts AI algorithms used for various smart city domains, and it analysed the future work based on collaborating Artificial intelligence with Big data along with virtual reality, 3D printing, etc. This work concluded with effective algorithms such as ANN, RNN/LSTM, and SVM. Data analytics can be used to forecast patterns in water quality in the future. Water quality managers can use this information to prepare for possible issues and implement preventative measures. Data analytics, for instance, can be used to forecast the effects of climate change on the standard of the water. (Kalini, 2017) Using this knowledge, measures can be created to safeguard water quality in the face of climate change and geographical analysis. Spatial data, such as where water bodies are located and where pollutants are distributed, can be analyzed using data analytics. This can assist in locating regions most vulnerable to poor water quality. Data analytics, for instance, can be used to visualize how pollutants are distributed throughout a watershed. The places that require pollution control measures the most can be determined using this information. Data analytics can be used to create machine learning models that can find trends in data on water quality. These models can be used to forecast future water quality trends and spot potential issues. (Sn et al., 2020) Machine learning algorithms, for instance, can be used to find trends in water quality data that are connected to certain pollutants. Using this data, early warning systems can be created to notify managers of water quality and potential issues. These are just a handful of the numerous ways that data analytics can support water quality monitoring. Water quality managers can better identify trends and patterns in water quality by employing data analytics. This information can be employed to control water quality more effectively.

(Yang et al., 2019) Based on the support vector machine frequency data, it can be forecasted in short-term high-frequency data. (Duan et al., 2019) reviews the various advancements in AI algorithms to improve their computation of storage and power. (Kalini, 2017) (Sn et al., 2020) I.J.I.M.'s paper used the AI research tool to analyze the various data mining methods.

Employing IoT and machine learning for water quality monitoring offers additional capabilities for collecting, analyzing, and interpreting data using data analytics.

a) IoT for Data collection: IoT devices like sensors and probes are deployed in water bodies. The collected data is transmitted

Table 5
Analysis of Machine Learning Algorithms implemented for assessing water quality.

Sl. No.	Pilot scale experiments	Water quality parameters	Machine learning techniques Proposed	Range Achieved	Preferred algorithm	Reference
1	1. Reclaimed water 2. Portable water	pH, Cl, Conductivity, Temperature, Turbidity, C	C.O.D.1. Direct SVM 2. PCC-SVM-based method 3. Single parameter method	SVM: (DA 97.22 % T.P.R.values 91.67 % FF.P.R.being 0) PCC-SVM: (DA> 97 %, TT.P.R. 93 % and FF.P.R.= 0)	PCC-SVM based method	(Xu et al., 2019)
2	Water quality monitoring station of Guazhou Water Source of the Yangtze River	Water Temperature, pH, Dissolved Oxygen, C.O.D. Conductivity, Turbidity, NH3-N	1. L	L.S.T.M.2. AA.R.I.M.A.3. SS.V. R.Period length $m = 10$ LSTM-0.0017 ARIMA-0.0126 SVR-0.0046	L.S.T.M.	(Limb et. al, 2011)
3	packaged drinking water, Pipeline drinking water, and pond water	pH T	T.S.Reference Indicator based, Decision Tree, C4.5, .Logistic Regression	MAE-0.17 RME-0.20	Reference Indicator-based Decision Tree	(Shakhari & Banerjee, 2019)
4	Evaluated quality of the Ebinur Lake Watershed surface water	pH, BB.O.D. NH_3^+, Iron, Copper, Zinc, D.O. TTDS Cl−, Na, etc. (20 parameters)	WQImethod, DI, RI and NN.D.I. correlation coefficients, SVM model	R^2–0.92, RMSE-58.4, RPD-2.81	Particle swarm optimisation (PSO)-support vector regression model	(Wang et al., 2017)
5	A case study Tilesdit dam in Bouira (Algeria)	pH, EC, T, TU Mg, B, H	1. L.S.T.M.RNNs 2. Multi-class SVMs	PCA 95.17 %. LDA 97.84 %, ICA 95.42 %	LSTM RNN	(Dilmi & Ladjal, 2021)
6	Rawal watershed, situated in Pakistan	Temperature, turbidity, pH, T	TDSsupervised prediction algorithms	MAE =1.96, F1score=0.56 Recall=0.564	Regression Algorithms	(Ahmed et al., 2019)
7	GRACE ΔTWS image reconstruction	Full image prediction to determine groundwater quality prediction	Extreme gradient boosting, multivariate linear regression, random forests, multilayer perceptron, and support vector regression	MAESVR Mean absolute error-2.70, RMSE SVR Root Mean Square Error-5.72	support vector regression	(Hussein et al., 2020)
8	Four monitoring stations of Johor River.	Ammoniacal Nitrogen, suspended solid (SS.S., pH	Adaptive Neuro-Fuzzy Inference System, Radial Basis Function Neural Networks	pH, No. of neuron-18, RMSE-0.15 Maxerror%−3.22, R^2=0.9	Neuro-Fuzzy Inference System (WDT-ANFIS)	(Tiwari et al., 2018)
9	Monthly data from 2 water quality monitoring stations within the Talar catchment	B.O.D. C.O.D. D.O. pH, turbidity, electrical conductivity, and Total solids	Random forest, Random tree, Reduced error pruning tree, Around 12 hybrid data-mining algorithms	R2 = 0.941, RMSE = 2.71, MAE = 1.87, NSE = 0.941, PBIAS= 0.500	BA-RT	(Dohare et al., 2014)
10	Kaggle-Indian water quality data sets	D.O. pH Conductivity, BB.O.D. Nitrate, Fecal coliform, and total coliform	SVM, K-NN, Naive Bayes nonlinear autoregressive neural network N	N.A.R.N.E.T. LL.S.T.M. Accuracy-97.01 % Sensitivity-99.23 % Specificity-97.78% Precision-94.93 % R.N.A.R.N.E.T.96:17 % R.L.S.T.M. 4:21 %	SVM model (WW.Q.C. NARNET model (WQI)	(Uddin et al., 2021)

wirelessly to a central database or cloud platform, enabling remote monitoring and access to information.

b) Data Integration and Management: The data management system organizes, stores, and preprocesses the collected data.
c) Data Analysis using Machine Learning: Various algorithms in identifying the abnormal water quality or potential pollution sources. It also identifies patterns of water quality data and develops models for predicting future water quality conditions (Almetwally et al., 2020).
d) Prediction models: By training the models on historical data, the system can identify real-time notifications.
e) Spatial models and statistical analysis: Data analytics are used to identify the most affected areas in water quality, which is highly risk. Statistical analysis tends to identify the patterns in water quality areas.

By combining IoT and machine learning, water quality monitoring systems can provide real-time insights, predictive capabilities, and proactive management approaches, which ultimately support the protection of water resources.

Scope for water quality analysis using geospatial technology, sensor technology, and machine learning

One of the main challenges related to collecting contaminated water is identifying the threats, analyzing water quality, and producing improved water resources management. Water is the most critical need of any living organism and essential for supporting the existence of all animals and individuals. There are definite restrictions on contaminations in most encompassing water bodies like waterways, lakes, and streams, which have explicit quality guidelines. e. The decay of water quality impacts oceanic biological systems by reducing the supply of safe new water for domestic use. Developing nations frequently go through quick financial expansion, and each improvement project has the possibility of producing adverse consequences on the climate. Recently, the abundant usage of manure has generally moved into groundwater streams. Moreover, the standards and specifications vary for different applications/uses. For instance, the water system is neither saline nor contains harmful materials, and it can be moved to plants or soil and accordingly annihilate the environment. In wireless sensor applications, Sondes furnished with sunlight-based chargers for energy harvesting is updated in recent works. Sondes are outfitted with IEEE 802.15.4-based remote connection points and mounted on floats. They structure a Remote Sensor Organization to speak with one another and send their estimations at standard stretches to a focal PC. through WSN in terms of GIS implementation, the movement of water and contaminants may be tracked, the places at risk of pollution and it can be found using geospatial technologies. Sensor technologies can be used to continuously gather data and measure water quality indicators at specific places. Data on water quality can be analyzed using machine learning methods to spot trends and patterns. Additionally, machine learning can be used to forecast changes in water quality, and this can be helpful in developing mitigation strategies and locations that are most likely to be impacted by pollution. Processes affecting water quality can be modeled using geospatial technology, sensor technology, and machine learning methods. We should understand how contaminants affect water quality and develop plans to enhance water quality standards.

description for Fig. 7 is given below the figure.

Fig. 7 shows the WQI values given for the groundwater tests - G7 and G21. The spatial guides of the review region are obviously found in the places with the most extreme defilement and leachate development from the landfill. The WQI standards are related to levels based on the threshold level or weightage assigned, and GGIS software is used to analyze the water quality in a specific area. It is GIS, a robust storage and data analysis tool. Using GIS tool, effective water potential zones in a particular area and targeted sites with hydrological surveys are analyzed, and the study was conducted by fixing the longitude and latitude points of the specific location. The government and water strategy policies can utilize this information as a solid fundamental reference in choosing reasonable destinations for groundwater assets. The mix of remote detecting with geographic data frameworks (GISs as a flexible device has expanded the exactness of potential groundwater zones. The selection of efficient sensors and processing units is required to get useful information, and these parameters are sound communication systems for transferring collected datasets without any loss or deviation. Table 5 shows the outcomes with optimal features of various models of deep learning techniques. The irrigational water quality indices values are revealed, and 33 % of the samples were not encouraged for irrigational purposes mentioned in this study. The most accurate model was predicted using ANN-3F with RMSE for Training and Testing coefficients as 0.002 and 0.004 and R^2 values 0.999 and 0.999.

From Table 6, two metrics, R^2 and RMSE values, are used to evaluate the performance in predicting the water quality index. If the RR.M.S.E. value is low, it indicates that studies have been directed to look at the performance of machine learning and A.I. techniques for WQI estimation. These examinations have found that deep learning strategies and ANN have outperformed traditional machine learning methods, such as multivariate regression. The model prediction is close to actual WQI values, and these models can be used to monitor the water quality and identify the areas where water quality is declining. Implementation of WQI with a machine learning algorithm is imperative to analyze deviation levels accurately. Big data analysis needs data mining for training and testing by the different algorithms. The big training data sets need an efficient algorithm model, and the outcome should be increased at the lowest cost.

From Fig. 8, it is understood that the various studies proposed learning strategies that can be utilized to track down the ROC. capabilities with a severe level of precision (Aldhyani et al., 2020). These strategies can be utilized to investigate the way of behaving capabilities that makes the capabilities diverge. Fig. 9 shows Various performance evaluations of Artificial Intelligence to measure water quality management.

The articles included survey articles, peer-reviewed Journals, and Conference proceedings mentioned in Fig. 10. We confined further to investigate areas and shortlisted 249 articles. On this set, we led a bibliometric investigation. Then, at that point, we shortlisted 63 pieces for an orderly writing survey. We likewise distinguish the correlative or cross-areas utilized in the articles. Table 7 outlines a systematic classification of research queries to steer forthcoming inquiries into responsible water quality prediction evolution using Sensor technology, IoT, AI, GIS. These questions fall into three overarching categories: AI capability Development and Maturity, Responsible AI implementation and Management, and Research Methodology and Validation. Table 7 serves as a navigational guide, suggesting potential avenues for future research in Responsible Technologies by furnishing instances of research questions. (Akbarighatar et al., 2023)

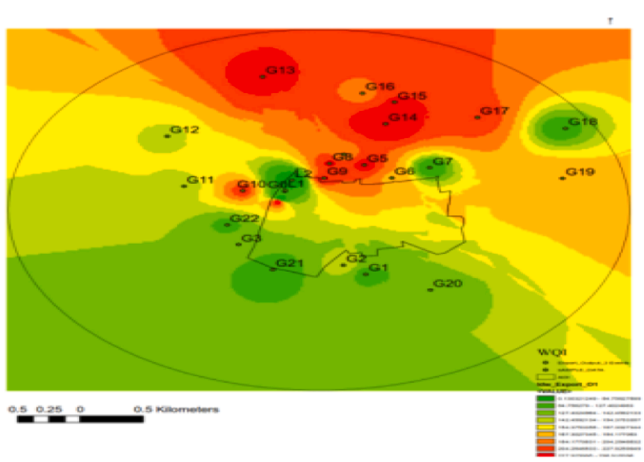

Fig. 7. WQI maps plot using GIS (Dawood et al., 2022).

Table 6
Analysis of Machine Learning Algorithms implemented for assessing water quality (Wang et al., 2017) (Sanya et al., 2022).

Variables	Model	Training		Validation	
		R^2	R	R.M.S.E. R^2	RMSE
IWQI	Gradient Boosting Regression	0.991	1.481	0.951	2.526
IWQI	Artificial Neural network (ANN)	0.975	2.482	0.958	2.172
WQI	Bayesian models	0.85	0.56	0.8	0.65
WQI	GA-BPNN	0.932	3.0	0.976	3.13
WQI	RF	0.96	1.37	0.93	3.1
WQI	GEP(Gene expression programming)	0.95	14.4	0.89	12.74
WQI	Ensemble models	0.9	0.4	0.8	0.5

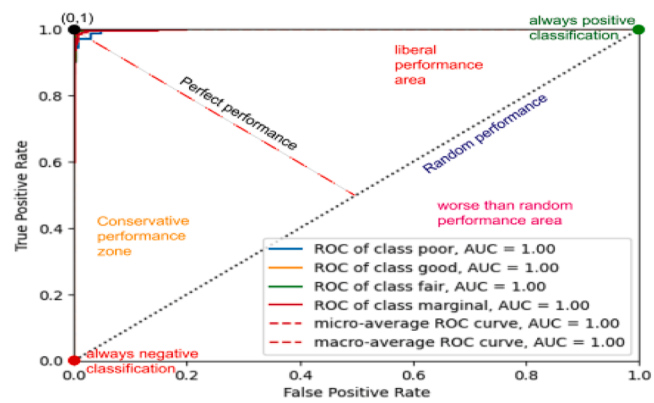

Fig. 8. Region of convergence curve to determine the water quality class.

Conclusions

Water is one of the essential natural resources throughout the world. Various physio-chemical and biological parameters determine the excellent quality of water. WWQI is calculated using the standard equation along with tedious laboratory observation. The machine learning techniques are applied to reduce the uncertainty of the aggregating process, robustness, and efficiency and lead to water quality analysis quickly and accurately at a cheaper rate. IoT-based real-time monitoring data from the various multiple sensors are collected for the performance analysis of Artificial Intelligence techniques. Water parameters are identified and validated using big data analysis, and IoT architecture in water distribution systems gathered from Integrated sensors is communicated to the cloud for the effective database. The information collected from various review papers shows that practical machine learning algorithms are pretty helpful for a better water quality monitoring system. Machine learning algorithms such as (classification and regression models) are suggested for the improvement in the control of water quality. The spatial distributions can estimate the detection and

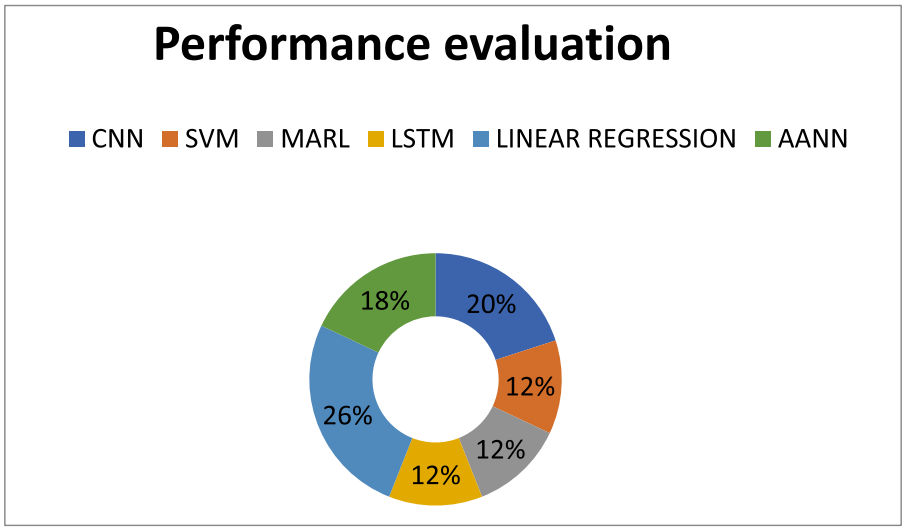

Fig. 9. Various Artificial Intelligence Methods to Analyze Water Quality Management.

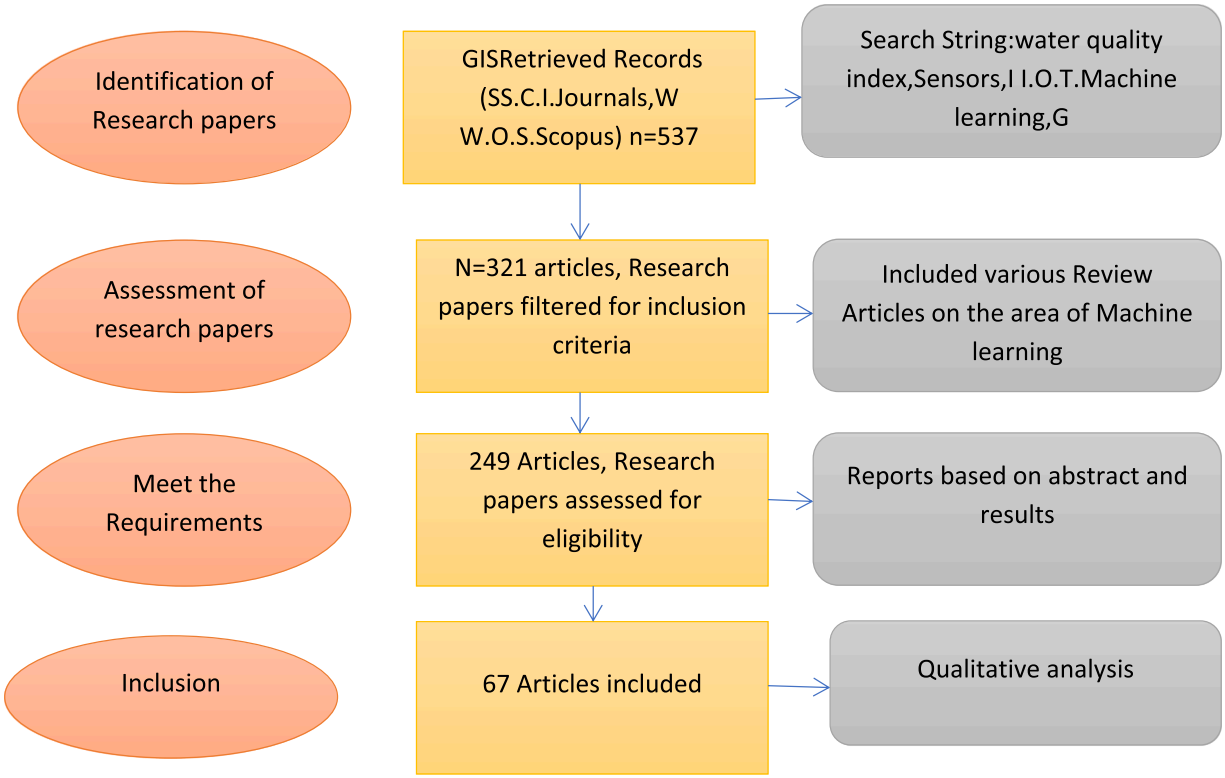

Fig. 10. Various Artificial Intelligence Methods to Analyze Water Quality Management.

prediction of water resources and their quality. Various modeling uses GRACE, Visual MODFLOW., and ArcGIS 10.2.2, in which the water zones and multiple parameters are identified and predicted. The effective and practical system for evaluating the nature of groundwater and its advancement in the utilization of physicochemical boundaries are appropriately upheld by GIS methods, multivariate displaying, and machine learning. Analytical Hierarchy Process is a central geological technique for identifying ground-level elevation levels, and they performed better boundary conditions and calibration levels. Using these models, we could ensure support for the sustainable management of water resources. Overall, numerical outcomes of predicting the water quality using WQI sensor technology, IoT, machine learning, deep learning, and GIS technologies are encouraging. These technologies can be utilized to create precise water quality prediction models, according to several studies conducted in the past. WQI of the Tigris River in Iraq was predicted using an artificial neural network. This model's R^2 and RMSE values are 0.91 and 0.27, respectively, in the traditional methods of machine learning, and it is. 0.94 and 0.15 in the random forest model. to forecast WQI most of the models achieved an R^2 value of 0.93 and RMSE value of 0.17.

CRediT authorship contribution statement

Poornima Jayaraman: . **Kothalam Krishnan Nagarajan:**

Table 7
Example of research questions for future work.

S. no	Domain Research Areas	Emerging Research Questions
1	Water Quality Index and Prediction	i) What strategies can be employed to create more inclusive Water Quality Index (WWQI models that account for diverse water quality parameters and the specific attributes of various water bodies? ii) How can we enhance the precision and dependability of WWQIcomputations, especially when dealing with unconventional water sources like wastewater and recycled water? iii)How can we adeptly convey WWQIdata to the general public and stakeholders, fostering awareness and encouraging practices that promote water conservation?iv) How can we incorporate real-time data from sensor networks and IoT infrastructure into predictive models to improve the timeliness and accuracy of water quality forecasts?
2	Sensor Technology	i)What approaches can be taken to create sensors with heightened sensitivity, selectivity, and durability capable of detecting an extended array of water quality parameters, encompassing emerging contaminants and trace chemicals? ii) In what ways can we advance the energy efficiency and lifespan of sensors to mitigate environmental consequences and lower maintenance expenditures? iii)How can we design sensor networks that are scalable, economically viable, and adaptable to a variety of water environments, including remote and challenging locations?
3	Machine Learning	i)What strategies can be employed to devise groundbreaking machine learning algorithms to manage extensive and intricate water quality datasets, accommodating challenges like missing data, outliers, and nonlinear relationships? ii) How can machine learning be seamlessly integrated with other data analysis techniques, like statistical modeling and data mining, to enrich comprehension of water quality patterns and trends? iii)How can we fashion machine learning models that offer Transparency in the decision-making process and facilitate the interpretability of results?
4	GIS(Geographical information system)	i)What methodologies can be employed to advance GGIS tools and techniques, facilitating the seamless integration, analysis, and visualization of spatial data pertinent to water quality, encompassing aspects like land use, topography, and pollution sources? ii)How can GGIS be effectively harnessed to pinpoint hotspots of water pollution, potential sources of contamination, and regions susceptible to water quality degradation? iii)What approaches can be taken to formulate GIS-based decision-support systems tailored for water resource management, pollution control, and regulatory compliance?

Conceptualization, Data curation, Formal analysis, Supervision, Writing – review & editing. **Pachaivannan Partheeban:** Conceptualization, Data curation, Formal analysis, Methodology, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Vallidevi Krishnamurthy:** Data curation, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jjimei.2023.100210.

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